

REVIEW

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Analysis of the current state of sewage sludge treatment from the perspective of current European directives

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Abstract

Due to its produced quantity, sewage sludge is a serious environmental problem not only in Slovakia, but also on a global scale. The submitted study deals with the current state of sewage sludge treatment with a focus on assessing the advantages and disadvantages of selected technologies. The main objective of the review study is to evaluate the advantages and limitations of individual sewage sludge processing technologies. Based on the evaluation of the advantages and limitations, another objective of the study was to determine the most suitable, less suitable, or unsuitable technologies for sewage sludge processing. Another objective of the study was to evaluate innovative technologies (not only commonly used ones). The purpose of this objective was to monitor the most current trends in sewage sludge processing technologies and, by evaluating them, to expand knowledge in this scientific field. The presented analysis of sewage sludge treatment was carried out in accordance with current European directives. The main novelty of the review study is a comprehensive assessment of sludge processing technologies with a focus on their main advantages and limitations. Another novelty is the focus of the study on innovative methods of sewage sludge processing. As part of the analysis, the evaluated sludge treatment technologies were divided into three categories: conventional, alternative, and innovative. As a conventional method, composting is currently the dominant method of sewage sludge management in the Slovak Republic. Mono-incineration, which is one of the conventional methods, allows for an effective reduction of the volume of sewage sludge while removing all unwanted hazardous components at the same time. However, this sludge treatment technology causes simultaneously the loss of most of the valuable nutrients, contained in sewage sludge. Pyrolysis and gasification as alternative technologies are effective in terms of removing hazardous components, but they are complex technologies, and their economic viability depends on their ability to maximise efficiency and produce high-value energy carriers. Hydrothermal liquefaction has a major advantage over all other thermal sewage sludge treatment processes because it does not require sludge drying. Innovative methods of sewage sludge treatment such as bio-electrochemical system, supercritical water oxidation, and biodiesel production require a more detailed examination by the scientific community to implement these methods in practice due to the lack of information from operational experiments. Given the extensiveness of the sludge treatment issue, we plan to address the economic aspect of the technologies in a separate study in the future.

Keywords Sludge, biogas, Process strategies, Innovative technologies, energy, Wastewater treatment plant, European directives

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Introduction

Sludge is increasingly a key problem in municipal and industrial wastewater treatment. It is generated as waste or a by-product of wastewater treatment processes, and the handling and disposal of this waste is complex and challenging [1]. Sludge from wastewater treatment plants (WWTPs) is an important by-product that contains important nutrients and organic substances. Sewage sludge is a by-product of wastewater treatment. It is rich in nitrogen (N), phosphorus (P) and potassium (K), but also contains harmful substances, including heavy metals and associated microorganisms [2, 3]. Sewage sludge is a by-product of the wastewater treatment process, which separates liquid and solid components [4].

The opinion of the European Committee of the Regions on the amendment of the Revision of the Urban Wastewater Treatment Directive C/2023/250 states that member states will take the necessary measures to ensure that sludge management methods are in accordance with the waste management hierarchy, in the Article 4 of the Directive 2008/98/EC. Such methods maximise the prevention of wasting the sources, reuse, recovery and recycling of sources, and minimise adverse effects on the environment. The Evaluation Report on Council Directive 86/278/EEC on the protection of the environment and soil when sewage sludge is used in agriculture was the source of data for this study. The Water Research Institute is commissioned to collect, verify, process and evaluate data on sludge produced from municipal wastewater treatment plants and methods of their management. This data and information shall be published on an annual basis.

Depending on where the sewage sludge comes from, it is divided into primary and secondary sludge. Primary sludge contains settling substances contained in wastewater (sludge from primary settling tanks). It usually has a granular structure, and dry matter ranges from 2 to 50 g/l. (the amount of solid component per total volume of the mixture). Secondary sludge (excess) is formed by a mixture of microorganisms and sedimentable substances from the biological stage of the wastewater treatment plants. This sludge is removed from the settling tanks, its coexistence is flaky, and the composition affects the method of treatment and the supplied wastewater. Primary and secondary sludge is referred to as crude. Crude sludge is still microbially active, it may contain pathogenic microorganisms, the total content of organic substances in the dry matter is around 70%. Stabilisation breaks down degradable organic substances, sludge can be pumped out and is considered harmless to health. Sludge stabilisation is the name for the process by which we adjust the final properties of the sludge so that spontaneous decomposition no longer occurs [5]. The aerobic

sludge stabilisation process is the most commonly used, in treatment plants of different sizes.

In most cases, it is a biochemical process of reducing the content of the residue of easily degradable organic substances and thus the total concentration of the organic component in the sludge. Determining the sludge stability criterion is not easy. In normal operating practice, the limit of stability of the sludge was considered to be 50% of the organic component, determined as the loss by annealing. Stabilised sludge must not decompose spontaneously, it must not burden the surroundings with odours. Simultaneously with the stabilisation of the sludge, the content of pathogenic microorganisms decreases, too, and the method of sludge stabilisation can also be a method of sludge sanitation [5].

The management of sewage sludge processing in Slovakia (Table 1) mainly includes application to agricultural land; application to forest land; composting and other recovery; energy recovery; incineration or landfilling.

Land application of sewage sludge is one of the important disposal alternatives [3]. Application of waste materials to agricultural land presents an opportunity for recovery of essential plant nutrients. Many waste products contain concentrations of plant nutrient elements sufficient to produce an agriculturally significant growth response [6]. The diverse composition of sewage sludge from different wastewater treatment plants and its seasonal variations require analysis of the chemical composition before its application to soil [3].

Forest sludge application offers many advantages. Its potential health risk is much lower than that of agricultural application, since forest products from berries and mushrooms to game contribute insignificantly to human nutrition, and forest soils and vegetation are often more amenable to sludge application. Given the perennial stands with extensive roots, year-round application is possible, weather permitting [7].

Composting is a complicated process aimed at destruction of pathogenic organisms; stabilisation of organic matter, ripening; drying of the sludge; production of material which can be environment ally used or sold [8]. Composting is a widespread method of treating organic solid waste (Bari and Koenig, 2001). Its performance is influenced by a number of factors, including the ratio of components in the mixture, aeration intensity, oxygen consumption rate, compost recycling, moisture level, pH value, C/N ratio, and others [9].

The composting process is used for sludge treatment and the transformation of complex biological waste into a stabilised product that can serve as an organic fertilizer and a value-added material. Composting occurs under aerobic conditions, where micro-aeration enhances the hydrolysis of complex substances into simpler

Table 1 Production of sludge from municipal WWTPs and its management from 2010 to 2022

Year	Amount of sludge (tonnes in dry matter)							
	Total	Recovered				Disposed of		Temporary storage in WWTPs
		Application to agricultural land	Application to forest land	Composting and other recovery*	Energy recovery	incineration	landfilling	
2010	54,760	923	0	47,140	0	0	16	6,681
2011	58,718	358	0	50,111	0	0	2,306	5,943
2012	58,706	1,254	0	46,446	3,196	0	1,615	6,195
2013	57,433	518	0	45,261	5,008	0	1,666	4,980
2014	56,883	8	0	36,524	16,038	0	1,073	3,240
2015	56,242	0	0	34,689	16,913	0	1,709	2,932
2016	53,054	0	0	34,695	10,975	68	2,359	4,957
2017	54,517	0	0	34,416	12,238	0	2,636	5,227
2018	55,929	0	0	32,982	11,677	0	2,451	8,819
2019	54,832	0	0	32,217	12,932	0	2,296	7,387
2020	55,519	0	0	36,562	11,928	0	2,302	4,727
2021	54,764	0	0	37,289	12,753	0	456	4,266
2022	55,049	0	0	33,509	10,326	0	1,540	9,674
Average	55,877	235	0	38,603	9,537	5	1,725	5,771

* Other method of recovery is reclamation of landfills, areas, production of growing substrates, etc.

compounds. This is achieved through the production of hydrolytic enzymes and an increase in the specific growth rate of microorganisms.

The composting process consists of three stages, classified based on temperature progression. In the first stage, mesophilic microorganisms thrive, causing a rise in system temperature. As the temperature continues to increase, the second stage begins, characterised by the activation of thermophilic microorganisms. During this phase, most pathogens are eliminated due to the high-temperature conditions. The final stage, known as the maturation phase, involves a gradual temperature decrease and the reactivation of the mesophilic microbial population [10].

Energy recovery from sludge is regarded as the most attractive methods for utilising the increasing quantity of sludge for eliminating volatile organic matter and reducing waste volume with possibility of recovering nutrients and/or metals. Most of these energy recovery technologies are not new as they have been developed commercially for other fuel types, however differences in the physical and chemical properties of sludge presents unique technical challenges which influences the energy recovery process and reactions [11].

Incineration uses an incinerator to heat and dry municipal sewage sludge, and then adopts high temperature to oxidise organic matters in municipal sewage sludge to achieve its reduction, stabilisation and harmless. Due to the limitation of land resource and the increment of

energy demand, incineration is considered as an effective and universal method to reduce the quantity and toxicity of MSS and to recover its energy [12]

Current sludge to landfill methods generally involve mixing the concentrated sludge with other solid waste in municipal landfills. However, stricter waste disposal legislation and higher landfill taxes are forcing the water industry to look for more efficient disposal strategies. Landfill operators are also increasingly reluctant to accept sludge material in the slurry state because of construction difficulties and the potential for instability of the landfill slopes [13].

Production of sludge in Slovakia from municipal WWTPs and its management from 2010 to 2022 is presented in Table 1.

Table 1 shows production of municipal wastewater sludge and its management from 2010 to 2022 in Slovakia. Data were obtained from Water Research Institute of Slovakia (WRI), which is committed to improving the handling, treatment and utilisation of sewage sludge in agriculture in order to eliminate its harmful effects on soil, water, plants, human and animal health, and to thereby ensure its correct and safe reuse. As required by Council Directive 86/278/EEC on the protection of the environment, and in particular of soil, when sewage sludge is used in agriculture (Sewage Sludge Directive), WRI verifies, processes and evaluates data on the sewage sludge produced in municipal sewage treatment plants and data on sludge handling methods in Slovakia. Such

data and information are published on an annual basis. The data itself comes from waste management reports that wastewater treatment plant operators are required to provide to regulatory authorities. In addition, EU member states are obliged to disclose annually the data listed in Article 10(1) of the Sewage Sludge Directive, in accordance with Article 10(2) of the mentioned directive (Council Directive 86/278/EEC of 12 June 1986 on the protection of the environment, and in particular of the soil, when sewage sludge is used in agriculture [14].

According to the latest Assessment Report, the predominant method of sludge stabilisation in the Slovak Republic is aerobic stabilisation, which is used by 74% of WWTPs. To a large extent, this stabilisation method is practised in WWTPs up to 10,000 PE. As a result, the production of aerobically stabilised sludge is only at the level of about 24% of the total sludge production. Approximately 26% of the total number of wastewater treatment plants use anaerobic stabilisation, of which part (about 3% of the total number of wastewater treatment plants in the Slovak Republic) uses Imhoff tanks. The mass fraction of anaerobically stabilised sludge accounts for approximately 76% of the total sludge production of the Slovak Republic. Chemical stabilisation of sludge (0.19% of sludge production) is carried out at three WWTPs. Sewage sludge has been confirmed in the

past as an effective adsorbent of heavy metals [15, 16] organic vapours [17] or tetracycline [18].

According to the statistics of the Water Research Institute, approximately 50,000 to 60,000 tonnes (in dry matter) of sludge has been produced annually at Slovak municipal wastewater treatment plants for the last 20 years.

Therefore, the technologies used to treat sewage sludge must be chosen with an emphasis on the ability to recycle the required components while effectively removing potentially hazardous substances for the environment and human health. Currently, there are several methods by which sewage sludge can be recovered or disposed of. Sewage sludge treatment is still a very current topic and requires a comprehensive view in the scientific field.

Sewage sludge utilisation is a critical aspect of wastewater treatment, with various methods being established worldwide. Figure 1 shows an analysis examines the trends in these methods from 2011 to 2022, focusing on agricultural use, composting, landfilling, incineration, and other unspecified methods according to Eurostat data. The data include reports from EU-27 countries, Iceland, Liechtenstein, Norway, Switzerland, United Kingdom, Bosnia and Herzegovina, Montenegro, North Macedonia, Albania, Serbia, and Türkiye, respectively [19].

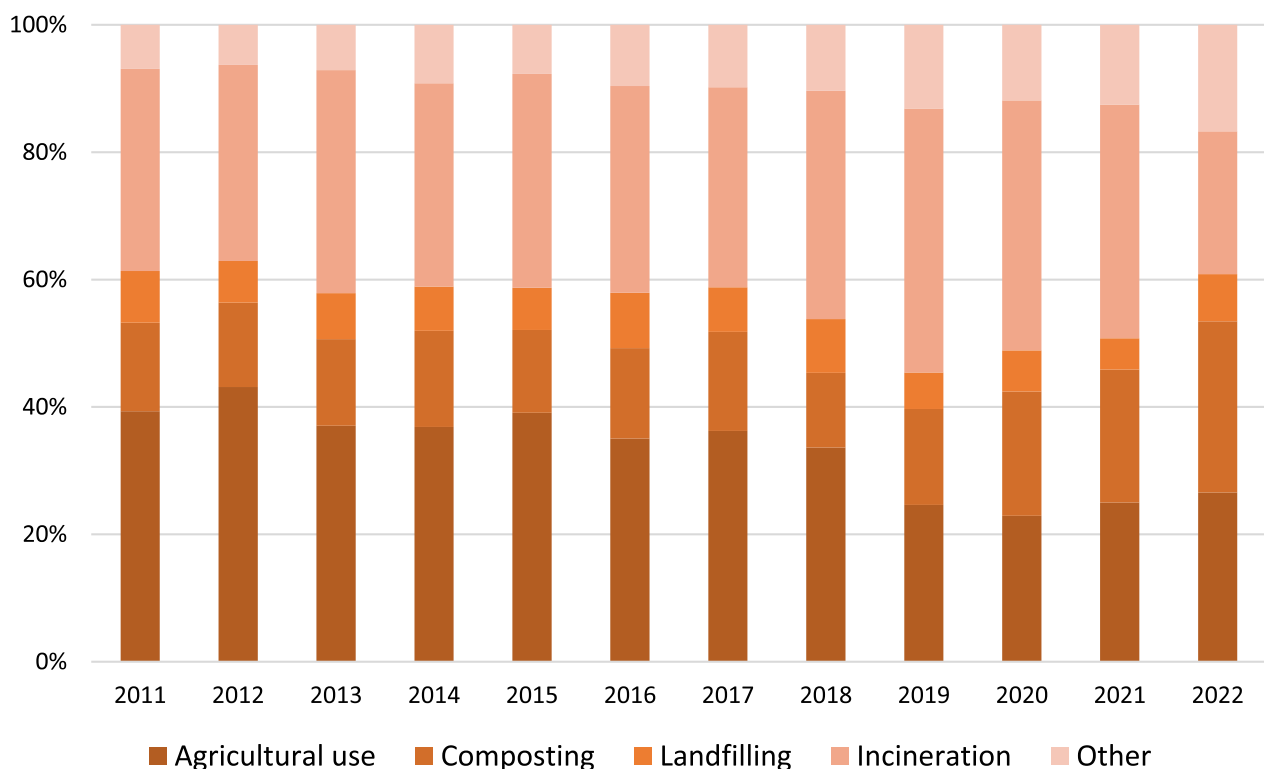


Fig. 1 Sewage sludge treatment methods in selected countries

The agricultural application of sewage sludge was the predominant utilisation method in the early years of the observed period. However, fluctuations occurred between 2011 and 2018, followed by a sharp decline from 2019 onward. By 2022, the volume of sludge used in agriculture had decreased significantly, likely due to stricter EU regulations on contaminants in sludge applied to farmland. Composting remained relatively stable over the years, with a slight peak around 2020, indicating a temporary increase in composting efforts. However, the overall trend does not suggest significant growth, implying that composting has not emerged as a dominant alternative. Landfilling exhibited a clear downward trend, with volumes steadily decreasing over time. A particularly sharp decline after 2019 likely reflects EU policies aimed at reducing landfill dependency and promoting more sustainable waste management practices. Incineration increased from 2011 to 2018, indicating a growing preference for thermal treatment of sewage sludge. However, in 2022, a steep drop in incineration volumes was observed. This may be linked to changing regulations, increased energy costs, or shifts towards more sustainable utilisation methods. The category of other disposal methods remained relatively stable, with minor fluctuations. Unlike other categories, it did not exhibit any dramatic increases or decreases, suggesting that these methods have maintained a consistent role in sewage sludge treatment management.

The main objective of the review study is to evaluate the advantages and limitations of individual sewage sludge processing technologies. Based on the evaluation of the advantages and limitations, another objective of the study was to determine the most suitable, less suitable, or unsuitable technologies for sewage sludge processing. Another objective of the study was to evaluate innovative technologies (not only commonly used ones). The purpose of this objective was to monitor the most current trends in sewage sludge processing technologies and, by evaluating them, to expand knowledge in this scientific field. In addition to the above, the objective of the presented review study is to provide a tool for decision-making when choosing a suitable technology for future sewage sludge processing operations. Only a comprehensive study can help in choosing the appropriate technology for processing this type of waste in a specific situation. The analysis of the current state of sewage sludge processing was prepared in accordance with European directives.

Characteristics of sludge from municipal wastewater treatment plants

According to Act 442/2002 (on public water supply and public sewerage systems and on amendments to Act

No. 276/2001 Coll. on regulation in network sectors), public sewerage is understood as an operationally independent set of objects and facilities serving public needs for the collective discharge of municipal wastewater [20]. According to Act 364/2004 (Act on Water and on amendments to Act of the Slovak National Council No. 372/1990 Coll. on offences, as amended (Water Act)), municipal wastewater is water from residential areas containing predominantly sewage wastewater; it may contain industrial wastewater, infiltrated water and, in the case of a unified sewer network or a semi-separated sewer network, also water from surface runoff [21]. It follows from the above that, under certain conditions, it is possible to discharge industrial wastewater through the sewerage network with sewage. However, in our study, we focus on the processing of paper sludge exclusively from municipal sewage. Depending on factors such as the development and size of settlements, industrial activities, population characteristics, and sewage services, sewage sludge can contain several toxic substances, such as heavy metals. Therefore, their presence must be analysed, since some of these metals are essential in agriculture (such as Zn and Cu). At higher concentrations, metals such as nickel and cadmium can have toxic effects in the soil, and the low mobility of such metals increases their concentration in the soil. Heavy metals (i.e. As, Cr, Hg, Mo, Pb, and Se) affect human and animal health, plant growth [22]. An important aspect related to the beneficial or toxic effects of heavy metals contained in sewage sludge is the pH value, since it affects their solubility. In particular, acidic media can increase the solubility of heavy metals and make them dynamically toxic; therefore, there may be a high risk associated with the acidic pH range. In case of pharmaceutical products, such as disinfectants, cleaning products, pesticides, dyes, paints, preservatives, food additives, personal care products, and organic pollutants, concerns increased due to the public health risk [23]. The pathogens present in sewage sludge come from human excrement and are directly related to the diet and health state of the population. They can also come from animal sources whose excrement is discharged into the sewerage system (e.g. dog and cat faeces) or via carriers in the sewers, especially rodents [24]. These microorganisms, including faecal coliforms, *Escherichia coli*, *Campylobacter* and *Salmonella sp.*, and parasites or viruses pose a risk to human health. Their presence is high in sewage sludge, especially in countries with low and medium economic level of populations [25, 26]. Despite the many benefits of using sewage sludge in soil, there are still many issues that need to be addressed [27].

About 50 to 80% of the original wastewater pollution is concentrated in sludge. These substances, such as phosphorus, nitrogen and potassium, are important for

the circular economy. The possibility of returning these substances to the soil could contribute to more sustainable agricultural production. However, in addition to their beneficial composition in terms of nutrient content, it is important that sewage sludge can also contain several harmful substances (e.g. pathogens, medicaments, microplastics, heavy metals, etc.). According to study [28] the range of major nutrients from eight studied WWTPs ranges for N $20,593 \pm 2265$ – $60,031 \pm 6603$ mg/kg; for P $10,663 \pm 521$ – $26,726 \pm 1098$ mg/kg; for K 4730 ± 198 – 8102 ± 315 mg/kg. [28]. Typical composition of sludge from municipal WWTPs is shown in Table 2.

Sewage sludge storage

Sludge can be stored under specific conditions to ensure environmental protection and compliance with regulations. In the European Union, the storage of sludge is primarily governed by the Sewage Sludge Directive (86/278/EEC) and other related regulations (EU Directive 86/278/EEC on Sewage Sludge. This directive regulates the use of sludge in agriculture and includes provisions for its storage to prevent risks to human health, soil, and water. According to the directive:

Storage must be managed safely: Sludge must be stored in a way that prevents environmental contamination, particularly to water sources. Time limits and conditions may apply: Some member states impose specific requirements on the duration and location of sludge storage. Quality control: Stored sludge must comply with heavy metal limits and pathogen reduction measures before application to land. National regulations: EU countries may impose stricter storage rules under their environmental laws.

Beyond Directive 86/278/EEC, the Industrial Emissions Directive (2010/75/EU) and the Waste Framework Directive (2008/98/EC) also apply to sludge storage in cases involving industrial waste or landfill regulations [14].

Identification of sewage sludge management methods

Each study has a separate treatment distribution key, but within our methodology, we decided to divide sewage sludge treatment technologies into conventional, alternative, and innovative methods. Thus, as is apparent from the foregoing information, the basic treatment of sewage sludge is to stabilise it in order not to cause damage to the environment and not to cause difficulties when being handled. However, the term stabilisation is not precisely defined by legislation and there is no universal criterion for assessing the stability of sludge.

The goal of sludge stabilisation is to reduce organic matter and thus reduce odour, destroy pathogenic organisms, improve drainage and reduce the volume and weight of sludge. In general, sludge stabilisation technologies can be divided into anaerobic and aerobic. Each of these methods has its advantages and disadvantages. Compared to aerobic stabilisation of sludge, anaerobic stabilisation generally has a lower energy performance, but it is more investment-intensive and is not suitable for smaller wastewater treatment plants, i.e. <35,000 PE [29]. Conventional, alternative, and innovative methods of sewage sludge treatment is presented in Table 3.

Bio-electrochemical system (BES) is considered as microbial fuel cell (MFC) that simultaneously generates bioenergy and treats wastewater or other biological waste. In this process, electroactive microorganisms on the anode break down organic matter, releasing protons and electrons. These charged particles then travel to the cathode via a proton exchange membrane and an electrical circuit, where they interact with an electron acceptor to produce electricity [62]. So far, only laboratory experiments are being conducted, as use has several challenges [63].

Supercritical water gasification (SCWG) is designed to produce hydrogen-rich gas from sewage sludge without any oxidant, whereas supercritical water oxidation

Table 2 Typical composition of sludge from municipal WWTPs

Component	Sewage sludge	References
Dry matter (%)	20–25	Wastewater treatment plants in Slovakia: Banská Štiavnica (January 2009/2012) Banská Bystrica (January 2009) Liptovský Mikuláš (October 2020) Matejovce (May 2024)
Organic components (% dry matter)	20–50	
Metals (mg/kg of dry matter):		
Cr	28/18 37 212 263	
Cu	280/240 460 1090 72	
Pb	29/120 5.7 26 9.6	
Ni	21/13 21 21 13.7	
Zn	4100/3500 940 748 593	
As	7.3/11 17 12.1	
Cd	21/35 1.6 < 1 0.49	

Table 3 Conventional, alternative, and innovative methods of sewage sludge treatment

Method	References
Conventional methods	
Application to soil	[30–32]
Mono-incineration/co-incineration	[12, 33, 34]
Anaerobic digestion	[35–37]
Landfilling	–
Alternative methods	
Pyrolysis	[38–40]
Gasification	[41–43]
Hydrothermal liquefaction	[44–46]
Reclamation purposes	[47, 48]
Composting	[49–51]
Drying	[52, 53]
Innovative methods	
Bio-electrochemical system	[54–56]
Supercritical water oxidation	[57–59]
Biodiesel production	[60, 61]

(SCWO) efficiently and environmentally friendly degrades organic compounds using an excess of oxidant [64, 65]. SCWO requires an excess of oxidant to enhance contact with organic compounds, which significantly drives up operating costs. In contrast, SCWG demands very high temperatures up to 1 027 K for complete gasification, leading to high capital expenses due to the harsh reaction conditions. Chematur Engineering AB (CEAB Corp.) of Sweden constructed Europe’s first pilot-scale plant for sewage sludge treatment, designed to process 7 m³/h. Since then, numerous pilot facilities have been built and operated across Europe, North America, China, and Russia, handling domestic and industrial sludge as well as hazardous industrial waste. For instance, the Harlingen Wastewater Treatment Plant in Texas operated a pilot-scale SCWO unit for sludge treatment, confirming the method’s viability. Additionally, a full-scale SCWO plant near Tianjin, China, has been successfully treating sludge at a wastewater treatment facility for the past 10 years [57].

Municipal sewage sludge is increasingly recognised worldwide as a valuable lipid feedstock for biodiesel production. This is because the sludge naturally contains high levels of lipids that are directly adsorbed to it. These lipids provide a significant source of energy, including triglycerides, diglycerides, monoglycerides, phospholipids, and free fatty acids from oils and fats. In addition, microorganisms consume organic and inorganic compounds as sources of energy, carbon, and nutrients during wastewater treatment. The cell membranes of these microorganisms—composed primarily of phospholipids—make up a

large portion of the sewage sludge, accounting for about 24% to 25% of the cell’s dry matter and providing approximately 7% of the oil from the dried secondary sludge. Additional studies have found that up to 36.8% by weight of the dry sludge consists of fatty acids and steroids, and since the fatty acids are predominantly in the C10 to C18 range, they are suitable for biodiesel production [66]. Although sewage sludge is readily available, its use in biofuel production is limited by several issues. These include costly pre-treatment processes, the requirement for significant amounts of organic solvents during lipid extraction, and the need to maintain product quality in line with established standards [67].

In Slovakia, but also in other countries of the world, waste management methods are carried out that are not suitable from an environmental point of view. The aim of this study was to provide an overview of methods for treatment of sewage sludge, the amount of which is growing worldwide. The methods were divided into the categories: conventional (commonly used), alternative (less used) and innovative (used on pilot basis or no use at all).

Material and methods

In our study, methods for the treatment of WWTP sludge are divided into conventional, alternative and innovative. In our study, we include the methods that are commonly used and economically available in Slovakia to the conventional ones. Alternative technologies are the technologies used to a lesser extent in Slovakia, but are still economically and technologically available and proven in practice.

Innovative methods are not used at all in Slovakia and are still less economically available, but from the environmental point of view they have many advantages. The comparison of sewage sludge treatment methods is focused on assessing the advantages, but also the limiting factors in the use of individual procedures. Overall, the following factors were used as evaluation criteria: financial cost, environmental factors, energy consumption, presence of hazardous substances and time consumption. This is because the new BREF documents (Best available techniques References documents) came into force during this period. For this reason, it is possible to assume a larger range of publications focused on the issue of sludge treatment.

The economic aspect is an important criterion for evaluating sludge treatment technology. In our study, less attention was paid to this aspect, because we focused more on criteria such as emission production, equipment life in the technology, hygiene risks, and the generation of difficult-to-use waste from the technology.

Innovative methods were searched using the databases *webofscience.com* and *scopus.com*. The search was conducted by entering a large number of keywords regarding treatment of sewage sludge and reading an enormous number of publications from the aforementioned databases.

Results and discussion

The division, description, and assessment of the methods, based on the above-mentioned criteria, are in Table 4.

Based on the processed References, the most suitable methods for the sewage sludge treatment are:

- the most optimal method in the combination of removing hazardous substances is mono-incineration;
- alternative thermal processes (pyrolysis and gasification);
- in all cases, they are the thermal methods that are the most effective in terms of the destruction of hazardous substances;
- pyrolysis and gasification due to high temperatures remove hazardous substances effectively. The output is a solid carbon residue, from which phosphorus can be recycled. However, it is necessary to add that these technologies are still in the research stage;
- innovative methods seem to be environmentally friendly, but they are not used in practice to a greater extent. The high cost of construction and operation is perceived as a negative factor of these methods. It is difficult to estimate how these processes, when being in full operation mode, will affect the environment in the future.

From the perspective of the expected future trends in sewage sludge management, we expect a trend of gradual tightening of the parameters of sewage sludge in terms of hazardous substances, which will disqualify its use in soil applications and, on the contrary, prefer thermal methods of sewage sludge management. Nowadays, some EU countries are introducing a legislative obligation to recover phosphorus from sewage sludge. Therefore, it is possible to assume that such an obligation will be implemented in the Slovak legislation in the near future.

The assessment of sewage sludge management methods requires an individual approach. When choosing treatment sludge management methods, it is necessary to find a compromise between the use of nutrients in the sludge and the effective removal of hazardous components. It was found out in the present study that the economic viewpoint should not be the only evaluation criterion. While direct exports of sewage sludge to agricultural land can be economically beneficial, there are legitimate

concerns about environmental and health risks. At the same time, phosphorus recycling is an important aspect for sustainable sewage sludge management techniques, as an extremely important element for fertiliser production, which is in acute shortage in Europe.

Based on the assessment, the identified methods of sewage sludge management from municipal wastewater treatment plants were divided into groups according to their suitability. The adaptability scale was determined based on the following criteria: possibilities for removing hazardous substances, recovery of nutrients, cost of technology, outputs from the technology and their use, energy demand, application in practice and legislation. This evaluation is presented in Table 5.

The methods listed as most appropriate are expensive and not suitable for small treatment plants. The methods identified as most appropriate are most effective, but in large facilities. Supplement with aerobic stabilisation.

In the case of the most suitable methods that were chosen, there are the thermal processes that were proven to be the most effective for the destruction of hazardous substances. These processes make it possible to use the energy contained in the organic matter of the sludge while minimising the risks associated with unwanted impurities in the sludge. Among these methods, mono-incineration is the most optimised, which at the same time ensures the highest phosphorus yield from ash after combustion. Sludge management, such as its direct application to the soil and landfilling, was identified as inappropriate regarding to the high risks of contamination and environmental threat, or in the case of landfilling, moreover, as the hierarchically most disadvantageous method of waste disposal without compensatory benefits.

Pyrolysis and gasification are other suitable thermal methods to remove hazardous substances contained in sewage sludge. In these processes, a carbon residue is produced, which allows the recycling of phosphorus. However, it should be noted that these techniques are not fully established in practice and their research is still ongoing. Composting is considered ineffective in removing microplastics and pharmaceuticals, while anaerobic fermentation brings only a partial reduction in the content of pharmaceuticals. In addition, compost and digestate/fugate containing sewage sludge are already limited by EU restrictions on CE certification of fertilizers, and there is a presumption of further tightening of the conditions for the application of such fertilizers to the soil or their strict ban. Although hydrothermal liquefaction is an alternative thermal process, it has a limited potential for phosphorus recycling due to maximising the yield of the liquid fraction—biofuel.

Innovative methods are environmentally suitable because they remove effectively heavy metals and

Table 4 Advantages and limitations of processing methods of sewage sludge

Processing method	
	Conventional methods
Direct application of sewage sludge to agricultural and forestry land	Technology description Only sludge that is treated and sanitised can be applied to agricultural and forestry land, because raw sludge contains a large amount of hazardous substances and microorganisms, which are harmful to both people and animals. The condition is that it undergoes anaerobic/aerobic stabilisation, stabilisation by liming or other stabilisation, or heat treatment that ensures sanitation or long-term storage The directive Directive 86/278/EEC stipulates that sewage sludge must be treated before being used as fertilizer. However, the use of untreated sewage sludge is permitted insofar when the sludge is washed down or buried in the soil. The directive furthermore stipulates that on pastures and fields used for forage cultivation, as well as during the vegetation period of fruit and vegetable crops, a waiting period prior to sewage sludge application must be observed [68] Advantages • If the sludge is uncontaminated and hygienically safe, it is the most accessible and cheapest method • Direct application of sewage sludge to the soil enables its maximum use as fertiliser and/or soil supplement; • An undemanding process in terms of technological requirements, expertise and finances; • No additional infrastructure is required to be built Limitations • Increased risk of the presence of pathogenic organisms and their transport from the sludge to animals and humans; • Increased risk of soil contamination with pharmaceuticals, heavy metals or microplastics; • Odour emissions due to insufficient stabilisation of the sludge before application to the soil; • Sludge can only be applied to the soil during a specified time-limited fertilisation period; • Considerable variability of the content of nutrients in sludge, which greatly complicates the management of nutrients in the soil in agricultural practice References [69] [70] [71] [72] [73] [74] [75] [76] [77]

Table 4 (continued)

Processing method	Conventional methods
Anaerobic fermentation	Technology description It is a method to stabilise sewage sludge. In the Slovak Republic, anaerobic stabilisation (fermentation) of sludge—methanisation was introduced. The final product is biogas and stabilised sludge (digestate). This process can take place either directly in WWTPs (in medium and large WWTPs for common stabilisation of primary and excess sludge, which are often thickened separately) or in external facilities, most often in biogas plants. Methanation is the most widespread sludge stabilisation process in the EU, using around 40% of the organic matter that is in raw sewage sludge. It is important to note that stabilising sludge does not necessarily mean its sanitation, i.e. pathogenic microorganisms can still occur in over the limit amount even in stabilised sludge, [5] The production of biogas depends primarily on the composition of the supplied sludge, during the stabilisation of which the biogas is released. It contains 60–70% methane—CH₄, 30–35% carbon dioxide—CO₂ and a small amount of water, sulphate—H₂S (approx. 0.1%), N₂, H₂, ammonia, fatty acids, etc. Thus, the main components of biogas are carbon dioxide and methane. Both of these gases are relatively difficult to separate from each other, and both are odourless (the smell of biogas is caused by sulphate). The production of biogas also depends on the retention time and temperature, i.e. factors determining the rate of decomposition, because a complete methanisation does not take place in the reactor, where a part of the sludge is continuously discharged [78]. Biogas volumes from sewage sludge can range from 0.75 to 1.12 m³ per kg VSS (volatile suspended solids), and the methane concentration is around 60% of the biogas composition [79] Pretreatments can be used to hydrolyse sludge and consequently improve biogas production, solids removal and sludge quality after anaerobic digestion. According to the nature of the technology, pre-treatment can be classified as thermal, physical (non-thermal), chemical and biological. There are two main thermal treatment/hydrolysis processes according to the applied temperature: high-temperature treatment (100–210 °C) and low-temperature treatment (< 100 °C) Physical processes for pre-treatment of sewage sludge are processes using high temperature, ultrasound, high pressure, electrical impulse and others, with a predominance of thermal and ultrasonic technologies. Chemical pre-treatments involve the use of chemical reagents to hydrolyse sludge (alkali pre-treatment, acid pre-treatment, ozonation. Biological pre-treatment processes aim to intensify hydrolysis during aerobic or anaerobic pre-steps or through the use of enzymes [80] Advantages • Production of biogas, which can be used for the production of heat and electricity in cogeneration, or for the production of biomethane; • A large reduction in the organic solids of the sludge; • Digestate is often largely free of pathogenic organisms (especially from thermophilic anaerobic fermentation, because it takes place at a higher temperature), has relatively little odour and can be used as fertilizer or soil conditioner to improve soil quality • Partial sanitation of sludge; • Enables connection with the pyrolysis or gasification process to increase the energy gain Limitations • High investment costs; • Long delay time of the process; • In order for the anaerobic fermentation process to be effective, sewage sludge must be mixed with other organic material; • Anaerobic fermentation by itself immobilises heavy metals only partially; • Anaerobic fermentation can partially reduce the amount of pharmaceuticals in sewage sludge, but it is important to add that this is only the initial stages of research into this issue; • Ineffective method in terms of removing microplastics; • The resulting product, which is applicable to agricultural land (digestate), is freed from organic matter by turning it into biogas • From July 2022, fertilizers containing digestates produced from sludge from municipal wastewater treatment plants cannot obtain CE certification and thus be sold on the EU market References [37, 78, 81, 82]

Table 4 (continued)

Processing method	Conventional methods
Mono-incineration	Technology description Nowadays thermal processes with sewage sludge can be divided into three basic groups, namely incineration (or so-called mono-incineration where a single type of waste is incinerated, which is the sewage sludge), co-incineration of the sludge with another raw material, fuel or waste, and alternative processes. The dry matter content and the organic matter content form the main combustible component, and therefore their proportions primarily determine the calorific value of the sludge. It is therefore advisable to dry it before incinerating. In most thermal processes of sewage sludge, they are usually partially dehydrated (85% of dry matter) or completely dehydrated (> 85% of dry matter) [83]. Dry sewage sludge is characterised by a calorific value in the range from 12 to 20 MJ/kg, i.e. a calorific value is lower than that of coal (14.6 to 26.7 MJ/kg), but equivalent to, e.g. the calorific value of lignite (11.7 to 15.8 MJ/kg) [84]. However, it should be added that after digestion or stabilisation, the content of organic substances in the sludge decreases and thus its calorific value decreases to values from approximately 8 to 13 MJ/kg [85]. Mono-incinerators of sewage sludge operate at a temperature in the range from 850 to 950 °C. Temperatures below 850 °C can lead to odour emissions, while temperatures above 950 °C risk ash sintering. The temperature of the combustion process depends on the energy content, the amount of sewage sludge processed and the amount of combustion air. On the example of German mono-incinerators, it can be stated that they apply sewage sludge either in its raw state or already digested sludge into the process. This can then be additionally drained, semi-dried or dried. In recent years, the stationary fluidised bed has prevailed in mono-incinerators as the most suitable type of sewage sludge combustion system [86]. According to Best Available Techniques (BAT) References Document for Waste Incineration—Industrial Emissions Directive 2010/75/EU (Integrated Pollution Prevention and Control) [87], the following important factors must be taken into account when incinerating sewage sludge: – Dry matter content (usually ranges from 10 to 45%—this has a great impact on the combustion process); – If the sludge is digested or not; – Lime, limestone and other sludge conditioning components; – Composition of the sludge as primary, secondary, biosludge, etc.; – Odour problems, especially when dosing the sludge and in storage areas (TWG, TWG Comments on Draft 1 of Waste Incineration BREF, 2003; TWG, TWG Comments on Draft 2 on Waste Incineration BREF, 2004) Options for sludge pre-treatment for mono-incineration are dewatering sludge up to 35% DS; dewatering sludge up to 45% DS; dewatering sludge up to 45%, add a partially drying up to 70% DS; Dewatering sludge up to 45%, add a complete drying up to 90% DS [88] Advantages • Makes it possible to significantly reduce the weight and volume of waste; • Complete stabilisation and sanitation of sludge; • Destruction of practically all organic substances (including pathogens and pharmaceuticals) and plastics/microplastics; • Concentration of phosphorus in ash; • Combined production of utility heat and electricity; • Partial removal of volatile and semi-volatile heavy metals; • Complete removal of odour Limitations • Suitable only for larger WWTP operations or so-called sludge centres; • Demanding permitting process; • Sludge with high moisture during incineration causes problems such as ignition delay, differential temperature and reduced temperature; • We can only get phosphorus from the incinerated sludge, the other nutrients are lost during incineration; • Approximately 30% of solids remain in the ash after incineration References [89] [56] [90] [91] [92]

Table 4 (continued)

Processing method	
Co-incineration	Conventional methods
	Technology description – Sewage sludge can also be co-incinerated with other types of raw materials or waste, i.e. it is used as an admixture of fuel in the combustion process, e.g. (municipal) waste incinerators, power plants, heating plants, etc. For example, in the operation of coal-fired power plants, it was found out that the addition of sewage sludge in the range of up to 5% of weight of charge does not cause any major problems in the process of control or emissions. Sewage sludge can be co-incinerated in power plants incinerating both brown and hard coal. The use of powder fuel or circulating fluidised bed are the main operating systems of the furnaces used for this purpose. In general, only stabilised (i.e. digested) sewage sludge is co-incinerated. Some power plants use only completely dried sludge, in others it is mixed with dewatered sludge and is added back to the incineration process [86]. The optimal proportion of sewage sludge co-incinerated in cement plants is around 7% of weight
Landfilling	Advantages • The permitting process may in certain circumstances be less demanding than in the case of a mono-incineration plant; • Co-incineration of partially or completely dried sewage sludge can save fossil fuels and thus costs and also CO₂ emissions; • The cement industry also uses ash from sewage sludge as a substitute for raw materials, which enables further cost reduction and helps save primary raw material resources Limitations • In existing facilities, it is necessary to solve the question of dosing sewage sludge, since it is not possible to use the paths of primary fuel to the incineration chamber for this; • During the co-incineration of sewage sludge, phosphorus-rich ash can be diluted or contaminated with pollutants or lost in other commodities; • Co-incineration of raw sludge with coal in power plants causes a high content of volatile substances such as mercury in air emissions; • Co-incineration of raw sludge with coal creates low-quality ash with limited use in construction as a building material; • Co-incineration of sewage sludge in waste incinerators creates increased emissions of dust particles in the waste gas References [12] [93] [94] Technology description Landfilling is the placement of waste in a landfill. It is a disposal activity. A landfill is a place with a waste disposal facility where waste is permanently deposited on or in the ground. An inert landfill is a landfill, where the waste producer disposes of their waste at the place of production, as well as a place that has been used for the temporary storage of waste, i.e. for more than one year. A landfill is not considered to be a facility or a place with a facility where waste is stored for the purpose of its preparation before its further transport to a place where it will be treated, recovered or disposed of, if the time of its disposal before its recovery or treatment does not usually exceed three years, or the time of its placement before its disposal does not exceed one year. If the sewage sludge is not used for another suitable purpose, it is disposed of by landfilling Advantages – Limitations • Limited capacity of landfills; • High costs, including transportation; • Uncompromising environmental standards; • Pressure to divert waste from landfilling Advantages –

Table 4 (continued)

Alternative methods					
Reclamation purposes	Technology description Reclamation is a type of landscape planning in which anthropogenically altered land is returned to its original state. Its main objective is to achieve a state as similar as possible to the original state and to mitigate environmental impacts by changing physical, chemical or biological properties. The effort is to create conditions for a self-sufficient ecosystem that is biologically and aesthetically acceptable. The most common objects of reclamation are landfills, contaminated areas, quarries, sandpits, and mines. Sludge is mostly used in the creation of reclamation substrates, where it is mixed with waste soil, fly ash from lignite combustion and other waste substances. The goal is to create a mixture that will be a suitable substrate for plant growth. This mixture is created by composting while checking its physical and chemical properties—moisture, nutrient content, pH. The resulting mixture must meet the limits set for the application of waste substances to the soil. These include, e.g. the content of heavy metals, the negative occurrence of pathogenic organisms such as <i>Salmonella</i> and the occurrence of various unfavourable bacteria [95] The pre-treatment system after reclamation can consist of the following sequence: thickening—anaerobic digestion—dewatering—land application [96] Advantages • Low costs; • Simple technology; • Possible use as reclamation material in various applications Limitations • Quality control of reclamation substrates; • Variable demand; • Content of heavy metals; • Legislative regulation of waste permitted for backfilling References [97, 98] <tr><td>Composting</td><td> Technology description Composting is classified as an aerobic process, in which biological waste is decomposed by microorganisms in the presence of air. Most often, the composting process is provided in the form of heaps, where the material is periodically mechanically rotated to achieve an even distribution of organic materials and sufficient contact with air. It is optimal to keep the process temperature between approx. 55 and 65 °C and the humidity at approx. 35 to 65% so that decomposition takes place at an appropriate rate and pathogenic organisms are destroyed. This process often takes months, depending on the conditions [99] It is desirable to combine sewage sludge from wastewater treatment plants with other waste materials, because it does not usually have an optimal composition for composting on its own (e.g. it can have too high water content). Therefore, in order to produce a suitable material for composting, the sludge is often mixed with wood chips, straw or green waste Pre-sun-drying (reducing the initial moisture content from 82% to 77.7%) can be one of the methods for pre-treating sewage sludge before composting [100]. Other methods of sludge pre-treatment before composting include enzyme secreting bacterial pre-treatment [101]; microwave [101]; thermal pre-treatment; alkali pre-treatment [10] Advantages • Enables a significant reduction in sludge volume; • When correct procedures are followed, it significantly reduces pathogenic organisms and pharmaceuticals • Correctly produced sludge compost can be used as a fertilizer or soil conditioner (improvement of soil quality) • Technologically less demanding process (lower investment and operating costs) Limitations • Relatively long duration of the process; • It is optimal to mix sewage sludge with other organic material (e.g. with wood chips, straw, biowaste, etc.)—usually cannot compost it separately; • Composting by itself does not immobilise heavy metals; • The opinion currently prevailing in the scientific community is that composting does not degrade microplastics; • During composting, gas emissions (e.g. carbon dioxide or ammonia) are produced, which cannot be evaluated in any way, while this can also be associated with an unpleasant smell; • Even if this process is less technologically demanding, it is necessary to follow the correct operating procedures; • From July 2022, fertilizers containing composts made from sludge from municipal wastewater treatment plants cannot obtain CE certification and thus be sold on the EU market References [50, 102] <tr><td></td><td> Technology description </td></tr></td></tr>	Composting	Technology description Composting is classified as an aerobic process, in which biological waste is decomposed by microorganisms in the presence of air. 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	Technology description				

Table 4 (continued)

Drying	Alternative methods		
	Contact dryers The heat transfer medium in contact dryers is usually hot oil or steam. The medium is indirectly led to the sludge via contact surfaces. Thus, the principle of contact drying is in heat transfer as in a heat exchanger. Dryers using the principle of contact drying are mainly by type: disc, thin-layer		
	Conventional dryers Conventional dryers include the most drying systems used for sewage sludge. The principle of conventional drying is the direct exposure of sewage sludge to a stream of hot air (heating medium). In the case of use for sewage sludge, the most commonly used types of these dryers are: fluidised, rotary drum, and belt dryer. Other used types include pneumatic, box, tunnel, turbine, spray dryers, etc		
	Solar dryer Solar dryers often combine 2 to 3 types of heat transfer: radiation transfer, convection transfer, and contact transfer. The contact transfer is used in the case of the application of additional heating of the bottom of the greenhouse, where thermal energy can be obtained, e.g. from wastewater through a heat pump. From a theoretical point of view, a solar dryer could also be classified as a conventional dryer, because the sludge is basically dried by contact with a stream of hot air, which is driven by fans The performance of a solar dryer is directly proportional to the sum of solar radiation generated by evaporation. In the conditions of the Slovak Republic, 1 MWh per square meter per year can be considered. Residual biomass or ash can be mixed into the loose dried sludge. It is a controlled and organised process for high-quality organic KSP (potassium sulphate) fertilizer. Ash is not an entirely necessary ingredient, but on the other hand, it can have some benefits [103]		
Advantages		Limitations	References
• Dried sludge is a suitable raw material for the production of granular fertilizer; • Drying of sewage sludge mainly achieves its hygiene, reduction of quantity and the associated costs of transport and/or storage; • Drying significantly multiplies the calorific value of the sludge, so for the subsequent thermal utilisation of the sludge, drying is an absolutely necessary step		• In the conditions of Slovakia, solar drying has a significant limitation, which is highly likely not to allow the sanitation of the sludge, in the case of incorrect operation, anaerobic conditions and the formation of odours may occur; • Solar dryers require a relatively large area, and their operation depends on weather conditions that cannot be controlled	[52, 103, 104]

Table 4 (continued)

Alternative methods	
Pyrolysis	Technology description Pyrolysis usually treats sewage sludge at temperatures above 350 °C without access to oxygen in order to obtain so-called biochar (pyrolysis coal or solid carbon residue), pyrolysis oil (sometimes referred to as biooil or biofuel) and pyrolysis gas. Pyrolysis oil and gas are mostly used to generate energy. The solid carbon residue is in solid form and can be used in power plants, heating plants or in agriculture. The output product of pyrolysis depends on the temperature of the process, where the charring yield decreases as the temperature increases. Therefore, a longer retention time of the charge in a low-temperature reactor with slow heating rates promotes the production of carbon residue, while a shorter or longer retention time at high temperature promotes the production of pyrolysis oil and gas [11]. The main product of pyrolysis is generally pyrolysis oil, while most studies in the field of sewage sludge pyrolysis suggest energy applications of pyrolysis oil, mainly for fuel production, less for applications in the field of production of valuable chemical products. Solid carbon residue is usually the main by-product of pyrolysis of sewage sludge with the primary production of liquid—oil. Sludge pre-treatments prior to pyrolysis can include aerobic pre-treatment (lower concentration in naphthalene); anaerobic and aerobic digestion (lower overall recovery of semi-volatile compounds compared to sludge without pre-treatment) [105]. Advantages • Reduction of sludge volume and weight; • Destruction of pathogenic organisms and pharmaceuticals; • Destruction of microplastics (some of them, however, may remain in pyrolysis products); • Stabilisation of heavy metals from sludge; • Obtaining energy from pyrolysis products; • Carbon residue is a potentially suitable carrier of inorganic fertilizers; • A profit of 14 to 57.5% by weight of the batch can be achieved; • Coal from the pyrolysis is suitable for the production of adsorbents; • Odour removal Limitations • Significant financial and energy demands of the pyrolysis process; • Carbon residue contains little accessible nutrients for plants and organisms in the soil, which makes its use as fertilizer very impractical; • Formation of pyrolysis oil containing toxic substances; • The calorific value of the solid residue is low (approx. 5 MJ/kg); • Complex process and economic viability of the plant is dependent on the ability to maximise efficiency and produce high-value pyrolysis oil, gas and carbon residue References [40] [106] [107] [108] [109] [110] [91] [111] [112]

Table 4 (continued)

Alternative methods	
Gasification	Technology description Gasification is a thermal process during which the carbonaceous content of sewage sludge is converted into flammable gas and ash in the presence of a reactive atmosphere, air or steam. Gasification converts mainly organic materials into combustible gas (also referred to as "syngas"), using 20% to 40% of the stoichiometric oxygen needed for perfect combustion, while pyrolysis is a thermochemical reaction that takes place at increased temperatures (500 to 1000 °C) and theoretically in an inert atmosphere without access to oxygen. The gasification process uses heat, pressure or water vapour to convert solid organic substances into syngas, which is a mixture of carbon monoxide, hydrogen, and other gases and by-products, i.e. charred coal, or oils and water. The calorific value of syngas from sewage sludge is around 4 MJ/m³. Gasification is best if the sludge is dried above 90% of dry matter, but it can also take place with partially dewatered sludge (20 to 40% of dry matter). The gasification process takes place at temperatures of 650 to 900 °C. Typical sewage sludge gasification gas is characterised by a high hydrogen content of 8.89–11.17 (vol. %), CO (6.28–10.77 vol. %), CH₄ (1.26–2.09 vol. %), C₂S (0.75–1.2 vol. %), carbon dioxide and gasification medium. 7In the first stage of the process, the sewage sludge is converted into a flammable lean gas with a temperature of 850–880 °C, which, in addition to nitrogen, consists mainly of hydrogen, methane, carbon monoxide or oxide and long-chain hydrocarbons. In the second stage of the process, hydrocarbons are decomposed by partial oxidation into methane, carbon monoxide, and hydrogen. The resulting synthesis gas is then purified and incinerated, most often in cogeneration, and thus serves for the combined production of electricity and heat [112]. Torrefaction was confirmed as a suitable method for pre-treatment of paper sludge. Torrefaction, also known as mild pyrolysis, is a thermochemical conversion process that introduces a solid feedstock within an elevated temperature oxygen-limited environment and upgrades it to a more energy dense, storable product [113]. Advantages • Significant reduction in weight and volume of sludge; • Sub-stoichiometric oxygen content leads to lower emissions into the air; • The produced synthesis gas has a wide range of uses; • An integrated gasification system with fuel cells allows to achieve electrical efficiency at the level of 30%; • Slag from gasification is suitable for use in building materials; • The whole process is energetically self-sufficient; • Captured solid particles from cleaning are either recycled to the gasifier or processed for metal recovery Limitations • Large initial investments; • Complex multi-stage process; • Synthesis gas must be thoroughly purified; • Necessary external input of energy to start the reactions; • More fundamental research is needed on the effect of sludge of different composition during gasification in order to obtain synthesis gas of the required quality for the production of energy or other products References [114] [115] [112]

Table 4 (continued)

Alternative methods	
Hydrothermal liquefaction	Technology description Hydrothermal liquefaction is a thermal depolymerisation process that is used to convert humid biomass and other macromolecules into petroleum-like substances at moderate temperature and high pressure. This process is also called “water pyrolysis” [116]. It is a process in which biomass undergoes thermochemical decomposition in inert atmosphere and produces a liquid bioraw material at increased temperatures and heating rates with relatively short reaction times [117, 118]. Water is used as a reaction medium, which acts both as a solvent and a catalytic agent. The average yield of methyl esters from lipids contained in sewage sludge is 24%, and the total average of saturated fatty acids reaches up to 55%. However, extracting lipids from sludge is expensive and requires a large volume of organic solvent and acidic catalysts [119] Advantages • Drying of sewage sludge is not required and thus hydrothermal liquefaction is a suitable alternative to incineration; • An opportunity to process sewage sludge with a high water content; • Enables the production of biofuel without occupying agricultural land, such as, e.g. in the case of purposefully grown energy crops; • Potential use of carbon residue for the purposes of supercapacitors and electrodes production Limitations • Expensive process of obtaining biofuel due to the need for lipid extraction; • High consumption of organic solvents and acid catalysts; • Excessive amounts of oxygen-containing compounds in raw biofuel can cause low calorific value, instability and high corrosivity and viscosity; • the obtained biofuel has a relatively high content of nitrogen (0.3 to 8%) and sulphur (0.3 to 1.6%), which is not desirable from the point of view of the fuel; • Heavy metals, micropollutants and microplastics are present in biofuel References [45] [120] [121] [122]

Table 4 (continued)

Innovative methods	
Bio-electrochemical system	Technology description
	As a new technology, the BES enables sustainable wastewater treatment and energy recovery at the same time. MFC as BES is an environmental and ecological technology that obtains energy through the oxidation of organic substrates and is transformed into electricity using microorganisms as catalysts. They convert organic materials into electrical energy and collectively remove nitrogen and COD by preparing anaerobic and aerobic conditions [123] Microorganisms play a leading role in these systems as they produce electron-rich metabolites, maintain the redox gradient, generate redox mediators, and deliver electrons to the electrode through direct electrons transfer or through a soluble electron transfer mediator with the production of electric current as the end result [124]
	Advantages • Direct conversion of substrates energy into electricity (no need for co-generators); • In comparison to anaerobic digestion (AD), no need for further biogas treatments, e.g. desulfurisation, and avoiding methane emission into the atmosphere; • Efficient conversion of organics to energy; • Low excess activated sludge production; • Low dependence on environmental factors; • Feasibility also in places with insufficient electrical infrastructures; • Can recover nutrients like nitrogen and phosphorus from sludge; • These systems can also aid in the recovery of metals (e.g. copper, silver); • Can improve the stabilisation of sewage sludge Limitations • High costs—to make BES systems ready for application in the near future it is necessary to fabricate a low-cost cathode/cathode catalyst (heterogeneous) that demonstrates reliable performance in a circumneutral environment; • Low power density generation rate; • Maintaining stable operation over long periods can be challenging; • The electrodes and other components in BESs can degrade over time References [125–128] https://pubs.rsc.org/en/content/articlehtml/2023/su/d3su00112a

Table 4 (continued)

Innovative methods		
Supercritical water oxidation	Technology description	
	SCW treatment methods can be divided into SCWG, supercritical water partial oxidation (SCWPO) and SCWO technologies with increasing amounts of oxidants. Unlike normal water, supercritical water ($T > 374.1\text{ }^{\circ}\text{C}$, $P > 22.1\text{ MPa}$) has a number of unique properties, including a lower dielectric constant, a higher diffusion coefficient, and a lower viscosity. In addition, the solubility of organic matter and gases increases significantly in supercritical water [129]. The composition and yield of production gases are affected by many variables, including the properties of the input material such as humidity, process temperature, air-to-biomass ratio, and the presence of catalysts [130]. SCWO can be used for treatment of wet organic waste without drying prior to the process since water is the reaction medium. It allows for complete oxidation of organics, essentially without harmful emissions [131]	
	Advantages • The removal efficiency of organic matters after a SCWO process can reach up to 99.9% within just a few minutes; • Self-heating balance can be reached when the proportion of organic material is more than 3 wt % (weight percent); • Can eliminate the pre-drying procedure of conventional treatments and stabilise the heavy metals; • SCWG and SCWPO aim to generate hydrogen-rich gases from MSS using oxidant less than stoichiometric ratio; • SCWO is an efficient and environmentally benign method for completely degrading organic compounds using an oxidant excess; • By-products are water, carbon dioxide, and a small amount of inert solids	Limitations • High costs—high investment (the equipment cost for this SCWO plant is approximately 3.12 million dollars) and operating (The operating cost of this SCWO plant is approximately 110.65 dollars/tDS) costs; although heating self-sufficiency can be achieved in the commercial plant, considerable electric power is still needed to maintain the running of several pumps and stirrers • Slow heating rate is one of the major obstacles; • Heavy metals except Cr tend to transfer from initial biomass to liquid after a SCWG reaction; • Moderate corrosion of reactor; • While the process significantly reduces sludge volume, the remaining inert solids, although small in quantity, still need to be handled; • Shutdowns of reactor can also be energy-intensive and time-consuming
		References [115, 129, 130, 132]

Table 4 (continued)

Innovative methods	
Biodiesel production	Technology description
	The biodiesel production process using sewage sludge as a feedstock includes lipid extraction and esterification/transesterification or direct esterification/transesterification in situ. Sludge lipids are mainly derived from the absorbed and/or deposited lipids directly from domestic wastewater, phospholipids in cell membranes, microbial metabolites, and products of ruptured cells. Cellular lipids in sludge include ditriglycerides, triglycerides, phospholipids, monoglycerides, sterols, and free fatty acids. Phospholipids have a percentage of about 24–25% of cells based on dry matter. In addition, fatty acids, sterols and some aliphatic components make up to 36.8% of the composition of dry sludge. The focus of the production process includes the pre-treatment of sewage sludge for more efficient lipid extraction, the extraction of lipids from the sludge itself, the production of biodiesel using extracted lipid or directly from the sludge, and the production of by-products and other value-added products as part of biodiesel production [119]. Transesterification is considered the most suitable process. Therefore, it is used in most industries. The low cost, simplicity and reduced viscosity of the oil obtained in this way is suitable for engines and equipment. Transesterification is a process of converting triglyceride and alcohol into fatty acid alkyl ester and glycerol. The process of transesterification can be categorised as a catalytic and non-catalytic process [133]. The use of biodiesel has a remarkable impact on reducing global warming by reducing greenhouse gas emissions. It reduces the amount of particulate matter and also reduces smoke due to the presence of free soot [134]. Biodiesel produced from waste products or unwanted materials helps to deal with huge amount of waste that was collected in one way or another, e.g. sewage sludge [135–137].
	Advantages • Production of biofuel; • Leaching toxicity of the heavy metals in sludge residue meets relative regulation values; • Reusability of by-products (methanol, glycerol); • When the sludge residue after extraction of lipids used for digestion, the rate of methane production increased
	Limitations • High costs—important costs are the labour cost and the price of methanol and catalyst that must be added to the feedstock. The high value of vegetable oils as food products make the production of a cost-effective fuel very challenging; • Catalyst waste; • Sludge waste after the production; • Wastewater from washing of the crude biodiesel layer with potassium bicarbonate solution; • When the sludge residue after extraction of lipids used for digestion, the volume obviously reduced
	References [60, 61, 138, 139]

Table 4 (continued)

Innovative methods	
Thermophilic aerobic membrane treatments	Technology description The thermophilic aerobic membrane reactor (TAMR) has been used in treating sewage using oxygen to dismantle organic matter. Biological processes are accelerated, and retention time is reduced as temperatures range from 45 to 65 °C. Water treatment systems such as TAMRs are widely used in industrial and municipal settings. This process is often used as the primary step in a multi-stage treatment line because of the wide variety of reactor designs or using a single-stage process, an organic load can be removed by 97%. TAMR utilising up flow is one of the earliest designs well-defined in design and operation [140]. Thermophilic aerobic membrane reactor (TAMR) process combines the advantages of thermophilic and membrane treatments [141] Advantages • This technology has proven to be extremely stable and able to treat wastewater of different nature; • High level of COD removal (higher than 94%); • The effluents of thermophilic process are well biodegradable by a mesophilic biomass; • High degree of stabilisation of treated sludge can be achieved (VS TS – 1 = 30–40%); • Preliminary management costs analysis highlights the economic feasibility; • Faster kinetics of pollutants removal; • Low volumes required; • Low production of biomass Limitations • Critical conditions, especially oxygen supply interruption, can affect the performance of the pilot plant; • The longer non-aerated phase involved a significant reduction (40%) of volatile suspended solids concentration in the biological reactor and the increase of 30% in foaming power; • Rapid pH increases and interruption of oxygen supply can cause deterioration of process performance and biomass quality References [142–145]
Upflow anaerobic sludge osmotic membrane bioreactor	Technology description A forward osmosis membrane replaced the traditional three-phase separator of the UASB technology [146]. In forward osmosis, water is transported across a semipermeable membrane from a solution of low osmotic pressure (feed solution), such as wastewater, to a solution of high osmotic pressure (draw solution). The diluted draw solution can later be re-concentrated and clean water can be generated by reverse osmosis or membrane distillation. This way, forward osmosis can be used to extract water from wastewater sources [147]. An osmosis membrane bioreactor is an innovative technology that combines traditional biological treatment and forward osmosis (FO) membrane technology and is highly effective for wastewater treatment [146, 148]. This technology has been gaining increasing popularity in wastewater treatment and reclamation due to its excellent product water quality, low fouling tendency and high fouling reversibility over conventional MBRs [149] Advantages • High level of COD, PO₄³⁻ and NH₄⁺ removal (95%); • Forward osmosis relies on osmotic pressure instead of hydraulic pressure and fouling generally appears less severe; • Due to the high rejection abilities of the FO membrane, FO-MBR-RO (forward osmosis–membrane bioreactor) produces higher-quality water in comparison to MF/UF MBR-RO and reduces fouling on the RO membrane; • Reducing membrane fouling could contribute further to the economic sustainability of the process Limitations • Salt accumulation; • Membrane Fouling; • To reduce salinity, it would be necessary to include another technology (e.g. microfiltration membrane) to remove excess salinity, as MF permeate can effectively reduce excess salt in the bioreactor References [146–151]

Table 5 Sewage sludge management methods based on their suitability

The most suitable	Mono-incineration of sewage sludge Alternative thermal processes (pyrolysis and gasification)
Less suitable	Co-incineration sewage sludge Composting Anaerobic fermentation Reclamation purposes Hydrothermal liquefaction Bio-electrochemical system Supercritical water oxidation Biodiesel production Thermophilic aerobic membrane treatments Upflow anaerobic sludge osmotic membrane bioreactor
Unsuitable	Direct application of sewage sludge to soil Sewage sludge landfilling

indirectly reduce emissions. Phosphorus is recycled only in the BES method, but in the other evaluated methods, nutrients are not recycled. In BES and biodiesel production methods, energy carriers in the form of electricity and fuel are also outputs. The production of biodiesel also produces by-products (methanol, glycerol) that can be further utilised. From an economic point of view, these methods are very financially demanding and are not used in practice.

Conclusion

A key criterion for the management methods of sewage sludge from municipal wastewater treatment plants is the ability to remove hazardous substances and recycle nutrients, in particular phosphorus, which is a scarce commodity in the EU for the fertiliser industry. It is sewage sludge that is becoming an important potential source of this element. Taking this criterion into account, mono-incineration appears to be a very suitable technology for treating sewage sludge. Mono-incineration enables an effective reduction in the volume of sludge while removing all undesirable hazardous components (including pathogens, pharmaceuticals and plastics/microplastics, partial removal of volatile and semi-volatile heavy metals). At the same time, this technology allows for the concentration of phosphorus in the form of ash, which allows for the reuse of this nutrient. It should be noted that, with the exception of phosphorus, other valuable nutrients are lost during mono-incineration that cannot be reused. When comparing mono-incineration with co-incineration in terms of nutrient recycling, co-incineration technology is not able to ensure phosphorus recycling. In addition to this negative aspect, increased emissions of dust particles in the waste gas were observed during co-incineration of sewage sludge in waste incineration

plants. The advantage of co-incineration technology is in saving fossil fuels and CO₂ emissions. However, even without phosphorus content, the ash obtained from co-incineration technology is a valuable substitute for input raw material in the cement industry. Pyrolysis and gasification are effective in terms of removing hazardous components, but they are complex technologies and their economic viability depends on their ability to maximise efficiency and produce high-value energy carriers reaching secondary fuel quality. Otherwise, pyrolysis and gasification plants operate in the incinerator or waste co-incineration mode. From the perspective of nutrient recycling, the pyrolysis output also forms a carbon residue, which is a potential carrier of inorganic fertilizers. However, this advantage is determined by the low accessibility of nutrients to plants and organisms in the soil. For the above reasons, the technologies mono-incineration of sewage sludge; alternative thermal processes (pyrolysis and gasification) and co-incineration sewage sludge were evaluated as the most suitable technologies for the treatment of sewage sludge. Composting combines the advantages of removing unwanted components (significantly reduces pathogenic organisms and pharmaceuticals) and at the same time, sludge compost can be used as a fertilizer or soil conditioner to improve soil quality. In addition to the above advantages, the composting process also has significant disadvantages (production of gaseous emissions, heavy metals are not immobilised). In addition, a significant disadvantage is the fact that from July 2022, fertilizers containing composts made from sludge from municipal wastewater treatment plants cannot obtain CE certification. Anaerobic fermentation, like composting, has the advantage of removing pathogenic components in the digestate. At the same time, this digestate can be used as a fertilizer or soil conditioner to improve soil quality, just like composting. Anaerobic fermentation has a disadvantage with composting—it is not able to ensure complete immobilisation of heavy metals. Anaerobic fermentation allows for the reduction of pollutants (pharmaceuticals) in sewage sludge—however, these findings are still in the early stages of research. For the above reasons, the composting and anaerobic fermentation technology was evaluated as less suitable. Sludge processing for reclamation purposes does not provide either the removal of undesirable substances or the recycling of nutrients. An important advantage of this technology is the low price and simplicity of implementing the technology. When assessing these advantages or disadvantages, this sewage sludge processing technology was evaluated as less suitable compared to other technologies. Drying is a necessary pre-treatment step for several methods of managing sewage sludge, while solar drying has significant limitations in Slovak conditions.

Belt drying appears to be the most suitable solution among the evaluated sludge drying methods. Hydrothermal liquefaction has a great advantage over other evaluated thermal sludge management processes—it does not require sludge drying, which is energy-intensive. The process allows the production of biofuel without taking up agricultural land for purposefully grown energy crops. However, biofuel requires major modifications to meet the quality requirements for transport. Another specific advantage of this alternative technology is the potential use of carbon residues for the production of supercapacitors and electrodes. Bio-electrochemical system allows the recovery of nutrients and heavy metals from sludge. SCWO provides complete degradation of organic compounds using excess oxidants. However, the disadvantage of both of the above innovative technologies lies in their high cost. Although the production of biodiesel from sludge as an innovative technology has significant advantages (the leaching toxicity of heavy metals in sludge residues meets the relative regulatory values or the reusability of by-products), even in this case the high price is still an important disadvantage. For the above reason, all three innovative technologies were classified as less suitable. Innovative technologies such as thermophilic aerobic membrane treatments and upflow anaerobic sludge osmotic membrane bioreactor are connected by high COD removal levels (more than 90%). However, both technologies are still poorly investigated and therefore the question remains how effective these technologies will be in removing other pollutants (pharmaceuticals and heavy metals) as well as in recycling nutrients (such as phosphorus). For this reason, these two innovative technologies were also evaluated as less suitable. However, it is not excluded that in the future, when taking into account and researching further advantages of these innovative technologies, they will be included among the most suitable technologies for sewage sludge treatment. Disposal of sewage sludge in landfills is not sustainable from an economic or environmental point of view. An innovative method of sewage sludge treatment requires more attention from the scientific community when implementing it into the operational mode. These methods are very expensive and are not yet at the stage where they can be applied in the operational mode. When choosing sludge management methods, it is important to balance nutrient recovery with effective removal of hazardous substances. This study showed that economic factors alone should not be the only decision criterion. Future trends in sewage sludge management point to stricter limits on hazardous substances, likely prohibiting land use, and favouring thermal treatment. Moreover, as some EU countries mandate phosphorus recovery, similar regulations may soon appear in

Slovak law. The research results demonstrate significant potential for application in various aspects of sewage sludge management and treatment. In addition, a better understanding of innovative techniques to reduce environmental impacts was analysed, contributing to more sustainable waste management practices. As a result, the results of this study are expected to guide future policies and industrial practices and promote a more resilient and environmentally responsible approach to sewage sludge treatment.

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Author contributions

Conceptualization, J.S. and M.S.; methodology, M.S.; investigation, D.S.; statistical analysis, M.M.; writing—original draft preparation, J.P.; writing—review and editing, M.M.; visualization, J.P.; supervision, D.V. All authors reviewed the manuscript. All authors have agreed to the published version of the manuscript.

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Availability of data and materials

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable. Consent to participate was obtained from all individual participants included in the study.

Consent to publication

Consent to publish was obtained from all individual participants included in the study.

Competing interests

The authors declare no competing interests.

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